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A Proofs

This section collects all the proofs in this paper.

A.1 Technical lemmas

This section presents several technical lemmas that will be used in our proofs.

Lemma A.1 (c.f. Lemma D.2 in [69])**.** *If $x \in \mathbb{R}^{d\times 1}$ is Gaussian random vector of d dimension, mean zero and covariance matrix Λ, and $A \in \mathbb{R}^{d\times d}$ is a fixed matrix. Then*

$$\mathbb{E}[xx^\top Axx^\top] = \Lambda(A + A^\top)\Lambda + \mathrm{Tr}(A\Lambda)\Lambda.$$

Lemma A.2. *If $x \in \mathbb{R}^{d\times 1}$ is Gaussian random vector of d dimension, mean zero and covariance matrix Λ, and $A \in \mathbb{R}^{d\times d}$ is a fixed matrix. Then*

$$\mathbb{E}[x^\top Ax] = \mathrm{Tr}(A\Lambda).$$

Proof. Since

$$\mathbb{E}[x^\top Ax] = \mathbb{E}\Big[\sum_{i,j} x_i A_{i,j} x_j\Big] = \sum_{i,j} A_{i,j} \cdot \mathbb{E}[x_i x_j] = \sum_{i,j} A_{i,j} \cdot \Lambda_{i,j} = \sum_{i=1}^{d} (A\Lambda^\top)_{i,i} = \mathrm{Tr}(A\Lambda),$$

which completes the proof. □

Lemma A.3. *For any matrices $A \in \mathbb{R}^{n\times m}$ and $B \in \mathbb{R}^{m\times n}$, we have*

$$\mathrm{Tr}(AB) = \mathrm{Tr}(BA).$$

Proof. Since

$$\mathrm{Tr}(AB) = \sum_{i=1}^{n} (AB)_{i,i} = \sum_{i=1}^{n}\sum_{j=1}^{m} A_{i,j}B_{j,i} = \sum_{j=1}^{m}\sum_{i=1}^{n} B_{j,i}A_{i,j} = \sum_{j=1}^{m} (BA)_{j,j} = \mathrm{Tr}(BA),$$

which completes the proof. □

Lemma A.4 (From Lemma D.1 in [69]; Also in [39])**.** *Let $X \in \mathbb{R}^{n\times m}$ be a variable matrix and $A \in \mathbb{R}^{a\times n}$ and $B \in \mathbb{R}^{n\times m}$ be two fixed matrices. Then, we have*

$$\begin{aligned}
&\partial_X \mathrm{Tr}(BX^\top) = B \in \mathbb{R}^{n\times m},\\
&\partial_X \mathrm{Tr}(AXBX^\top) = (AXB + A^\top XB^\top) \in \mathbb{R}^{n\times m}.
\end{aligned}$$

Lemma A.5 (Von Neumann's Trace Inequality; Also in Lemma D.3 in [69])**.** *Let $A \in \mathbb{R}^{n\times m}$ and $B \in \mathbb{R}^{m\times n}$ be two matrices. Suppose $(\sigma_1(A), \cdots \sigma_{\min\{n,m\}}(A))$ and $(\sigma_1(B), \cdots \sigma_{\min\{n,m\}}(B))$ are all the singular values of A and B, respectively. We have*

$$\mathrm{Tr}(AB) \le \sum_{i=1}^{\min\{n,m\}} \sigma_i(A)\sigma_i(B) \le \sum_{i=1}^{\min\{n,m\}} \|A\|_2 \cdot \|B\|_2 = \min\{n,m\} \cdot \|A\|_2 \cdot \|B\|_2.$$

A.2 Proof of Proposition 1

This section presents the proof of Proposition 1.

Proof of Proposition 1. For the AT loss $\mathcal{L}(\theta)$ defined in Eq. (8), we have that

$$\begin{aligned}
\mathcal{L}^{\text{adv}}(\theta) &:= \mathcal{R}^{\text{adv}}(\theta, M_{\text{train}}) = \mathop{\mathbb{E}}_{\tau} \max_{\|\Delta_\tau^\top\|_{2,\infty}\le\epsilon} |\hat{y}_{q,\theta}(E^{\text{adv}}_{\tau,M_{\text{train}}}) - y_{\tau,q}|^2\\
&= \mathop{\mathbb{E}}_{\tau}\left\{ \max_{\|\Delta_\tau^\top\|_{2,\infty}\le\epsilon} \frac{1}{2}\left| \begin{pmatrix}(w_{21}^V)^\top & w_{22}^V\end{pmatrix} \cdot \frac{E^{\text{adv}}_{\tau,M_{\text{train}}} E^{\text{adv},\top}_{\tau,M_{\text{train}}}}{N + M_{\text{train}}} \cdot \begin{pmatrix} W_{11}^{KQ} \\ (w_{21}^{KQ})^\top \end{pmatrix} \cdot x_{\tau,q} - y_{\tau,q} \right|^2 \right\}. \quad \text{(A.1)}
\end{aligned}$$

Then, the term $E^{\text{adv}}_{\tau,M_{\text{train}}} E^{\text{adv},\top}_{\tau,M_{\text{train}}}$ can be decomposed as follows,

$$
\begin{aligned}
E^{\text{adv}}_{\tau,M_{\text{train}}} E^{\text{adv},\top}_{\tau,M_{\text{train}}} &= \left(\begin{pmatrix} X_\tau \\ Y_\tau \end{pmatrix} \quad \begin{pmatrix} X^{\text{sfx}}_\tau + \Delta_\tau \\ Y^{\text{sfx}}_\tau \end{pmatrix} \quad \begin{pmatrix} x_{\tau,q} \\ 0 \end{pmatrix} \right) \cdot \left(\begin{pmatrix} X_\tau \\ Y_\tau \end{pmatrix} \quad \begin{pmatrix} X^{\text{sfx}}_\tau + \Delta_\tau \\ Y^{\text{sfx}}_\tau \end{pmatrix} \quad \begin{pmatrix} x_{\tau,q} \\ 0 \end{pmatrix} \right)^\top \\
&= \begin{pmatrix} X_\tau & X^{\text{sfx}}_\tau & x_{\tau,q} \\ Y_\tau & Y^{\text{sfx}}_\tau & 0 \end{pmatrix} \begin{pmatrix} X_\tau & X^{\text{sfx}}_\tau & x_{\tau,q} \\ Y_\tau & Y^{\text{sfx}}_\tau & 0 \end{pmatrix}^\top + \begin{pmatrix} 0_{d\times N} & \Delta_\tau & 0_{d\times 1} \\ 0_{1\times N} & 0_{1\times M_{\text{train}}} & 0 \end{pmatrix} \begin{pmatrix} 0_{d\times N} & \Delta_\tau & 0_{d\times 1} \\ 0_{1\times N} & 0_{1\times M_{\text{train}}} & 0 \end{pmatrix}^\top \\
&\quad + \begin{pmatrix} X_\tau & X^{\text{sfx}}_\tau & x_{\tau,q} \\ Y_\tau & Y^{\text{sfx}}_\tau & 0 \end{pmatrix} \begin{pmatrix} 0_{d\times N} & \Delta_\tau & 0_{d\times 1} \\ 0_{1\times N} & 0_{1\times M_{\text{train}}} & 0 \end{pmatrix}^\top + \begin{pmatrix} 0_{d\times N} & \Delta_\tau & 0_{d\times 1} \\ 0_{1\times N} & 0_{1\times M_{\text{train}}} & 0 \end{pmatrix} \begin{pmatrix} X_\tau & X^{\text{sfx}}_\tau & x_{\tau,q} \\ Y_\tau & Y^{\text{sfx}}_\tau & 0 \end{pmatrix}^\top \\
&= \begin{pmatrix} X_\tau & X^{\text{sfx}}_\tau & x_{\tau,q} \\ Y_\tau & Y^{\text{sfx}}_\tau & 0 \end{pmatrix} \begin{pmatrix} X_\tau & X^{\text{sfx}}_\tau & x_{\tau,q} \\ Y_\tau & Y^{\text{sfx}}_\tau & 0 \end{pmatrix}^\top + \begin{pmatrix} \Delta_\tau \\ 0_{1\times M_{\text{train}}} \end{pmatrix} \begin{pmatrix} \Delta_\tau \\ 0_{1\times M_{\text{train}}} \end{pmatrix}^\top \\
&\quad + \begin{pmatrix} X^{\text{sfx}}_\tau \\ Y^{\text{sfx}}_\tau \end{pmatrix} \begin{pmatrix} \Delta_\tau \\ 0_{1\times M_{\text{train}}} \end{pmatrix}^\top + \begin{pmatrix} \Delta_\tau \\ 0_{1\times M_{\text{train}}} \end{pmatrix} \begin{pmatrix} X^{\text{sfx}}_\tau \\ Y^{\text{sfx}}_\tau \end{pmatrix}^\top,
\end{aligned}
$$

which further means that

$$
\begin{aligned}
&\left((w^V_{21})^\top \quad w^V_{22} \right) \cdot \frac{E^{\text{adv}}_{\tau,M_{\text{train}}} E^{\text{adv},\top}_{\tau,M_{\text{train}}}}{N + M_{\text{train}}} \cdot \begin{pmatrix} W^{KQ}_{11} \\ (w^{KQ}_{21})^\top \end{pmatrix} \cdot x_{\tau,q} \\
&= \left((w^V_{21})^\top \quad w^V_{22} \right) \cdot \frac{\begin{pmatrix} X_\tau & X^{\text{sfx}}_\tau & x_{\tau,q} \\ Y_\tau & Y^{\text{sfx}}_\tau & 0 \end{pmatrix} \begin{pmatrix} X_\tau & X^{\text{sfx}}_\tau & x_{\tau,q} \\ Y_\tau & Y^{\text{sfx}}_\tau & 0 \end{pmatrix}^\top}{N + M_{\text{train}}} \cdot \begin{pmatrix} W^{KQ}_{11} \\ (w^{KQ}_{21})^\top \end{pmatrix} \cdot x_{\tau,q} \\
&\quad + (w^V_{21})^\top \cdot \frac{\Delta_\tau \Delta_\tau^\top}{N + M_{\text{train}}} \cdot W^{KQ}_{11} x_{\tau,q} + \left((w^V_{21})^\top \quad w^V_{22} \right) \cdot \frac{\begin{pmatrix} X^{\text{sfx}}_\tau \\ Y^{\text{sfx}}_\tau \end{pmatrix} \Delta_\tau^\top}{N + M_{\text{train}}} \cdot W^{KQ}_{11} x_{\tau,q} \\
&\quad + (w^V_{21})^\top \cdot \frac{\Delta_\tau \begin{pmatrix} X^{\text{sfx}}_\tau \\ Y^{\text{sfx}}_\tau \end{pmatrix}^\top}{N + M_{\text{train}}} \cdot \begin{pmatrix} W^{KQ}_{11} \\ (w^{KQ}_{21})^\top \end{pmatrix} x_{\tau,q}.
\end{aligned} \tag{A.2}
$$

Inserting Eq. (A.2) into Eq. (A.1) and applying the inequality that $|a+b|^2 \le 2 \cdot (a^2 + b^2)$, $\mathcal{L}^{\text{adv}}(\theta)$ can thus be bounded as

$$
\begin{aligned}
\mathcal{L}^{\text{adv}}(\theta) \le\ & 2 \cdot \mathop{\mathbb{E}}_{\tau} \left[\left((w^V_{21})^\top \quad w^V_{22} \right) \cdot \frac{\begin{pmatrix} X_\tau & X^{\text{sfx}}_\tau & x_{\tau,q} \\ Y_\tau & Y^{\text{sfx}}_\tau & 0 \end{pmatrix} \begin{pmatrix} X_\tau & X^{\text{sfx}}_\tau & x_{\tau,q} \\ Y_\tau & Y^{\text{sfx}}_\tau & 0 \end{pmatrix}^\top}{N + M_{\text{train}}} \cdot \begin{pmatrix} W^{KQ}_{11} \\ (w^{KQ}_{21})^\top \end{pmatrix} \cdot x_{\tau,q} - y_{\tau,q} \right]^2 \\
& + \underbrace{2 \cdot \mathop{\mathbb{E}}_{\tau} \max_{\|\Delta_\tau^\top\|_{2,\infty} \le \epsilon} \left[(w^V_{21})^\top \cdot \frac{\Delta_\tau \Delta_\tau^\top}{N + M_{\text{train}}} \cdot W^{KQ}_{11} x_{\tau,q} \right]^2}_{:= A_1(\theta)} \\
& + \underbrace{2 \cdot \mathop{\mathbb{E}}_{\tau} \max_{\|\Delta_\tau^\top\|_{2,\infty} \le \epsilon} \left[\left((w^V_{21})^\top \quad w^V_{22} \right) \cdot \frac{\begin{pmatrix} X^{\text{sfx}}_\tau \\ Y^{\text{sfx}}_\tau \end{pmatrix} \Delta_\tau^\top}{N + M_{\text{train}}} \cdot W^{KQ}_{11} x_{\tau,q} \right]^2}_{:= A_2(\theta)} \\
& + \underbrace{2 \cdot \mathop{\mathbb{E}}_{\tau} \max_{\|\Delta_\tau^\top\|_{2,\infty} \le \epsilon} \left[(w^V_{21})^\top \cdot \frac{\Delta_\tau \begin{pmatrix} X^{\text{sfx}}_\tau \\ Y^{\text{sfx}}_\tau \end{pmatrix}^\top}{N + M_{\text{train}}} \cdot \begin{pmatrix} W^{KQ}_{11} \\ (w^{KQ}_{21})^\top \end{pmatrix} x_{\tau,q} \right]^2}_{:= A_3(\theta)}.
\end{aligned} \tag{A.3}
$$